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A Synopsis on

"INTELLIGENT CAREER PATH ADVISORY AGENT"

Submitted for the Major Project Work-Phase 1

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Declaration by Students

We, the undersigned students Deepanshu Bisht (1NT23CS056) and Divya S Karki (1NT23CS067) hereby declare that we are fully aware of the project conduction procedures and evaluation rubrics as prescribed by the department. We understand that the project work will be evaluated through continuous assessment across the defined phases.

Name of the Student	Signature
Deepanshu Bisht	
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Confirmation by Guide

I, Dr. Krishna Rao Venkatesh, hereby declare that I shall guide the above-mentioned students in the project entitled "INTELLIGENT CAREER PATH ADVISORY AGENT" and will ensure that the project work is carried out under my supervision and submitted as per the academic requirements for the academic year 2025–2026.

Signature of the Guide

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1. Introduction

In today's rapidly evolving job market, choosing the right career path has become one of the most critical and challenging decisions for students and early professionals. The proliferation of emerging technologies, shifting industry demands, and the increasing complexity of professional roles have made career planning a highly nuanced process. Traditional career counseling methods, which often rely on standardized assessments or generic advice, fall short in providing the kind of personalized, dynamic, and data-driven guidance that modern professionals require.

Despite the availability of various career platforms and job portals such as LinkedIn, Naukri, and Glassdoor, most existing systems focus primarily on job matching based on keywords and qualifications. They fail to consider the holistic profile of an individual, including their personality traits, soft skills, academic performance, extracurricular experiences, and long-term goals. Moreover, these systems lack the ability to explain their recommendations, leaving users uncertain about why a particular career path was suggested and how to bridge the gap between their current profile and desired roles.

This project proposes the development of an Intelligent Career Path Advisory Agent, a sophisticated AI-powered system that leverages Large Language Models (LLMs) and agentic reasoning to deliver personalized, explainable, and actionable career guidance. The system is designed to accept multi-dimensional user profiles as input, reason over real-time job market trends, and generate tailored career roadmaps backed by coherent explanations.

By combining the natural language understanding capabilities of LLMs with structured user profiling, retrieval-augmented generation (RAG), and multi-step agentic planning, the proposed system aims to fundamentally transform how individuals navigate their career journeys. This project bridges the critical gap between academic preparation and professional success, making intelligent career mentorship scalable, accessible, and highly personalized.

2. Objectives of the Project

The following specific and measurable objectives guide the development of the Intelligent Career Path Advisory Agent:

1. To design and implement a multi-dimensional user profiling system that captures academic performance, technical and soft skills, personality traits, interests, and career aspirations to form a comprehensive user model.
2. To integrate real-time job market data and industry trend analysis using retrieval-augmented generation (RAG) pipelines, ensuring that career recommendations are aligned with current professional demands.
3. To develop an LLM-powered agentic reasoning engine capable of decomposing complex career queries into sub-tasks, researching opportunities, and synthesizing coherent, explainable career roadmaps.
4. To identify skill gaps between a user's current profile and their target career roles, and provide structured, actionable learning pathways including relevant courses, certifications, and projects.
5. To build an intuitive conversational interface that enables users to interact with the advisory agent in natural language, ask follow-up questions, and refine their career plans iteratively.

3. Literature Survey and Review

A comprehensive review of existing literature was conducted across the areas of career recommendation systems, intelligent advisory agents, and the application of large language models in personalized guidance. The following summarizes the key findings and their relevance to this project.

Early career recommendation systems relied heavily on collaborative filtering and content-based filtering techniques, similar to those used in e-commerce recommendation engines. These approaches matched users to job profiles based on keyword overlap in resumes and job descriptions. While effective for simple matching, such systems lacked the ability to reason over complex user traits like personality, soft skills, or aspirational career transitions. Studies by Shalaby et al. (2017) and Qin et al. (2018) highlighted that purely skill-keyword-based matching results in low precision for non-linear career paths and career changers.

The emergence of transformer-based language models such as BERT and GPT marked a significant shift in how text-based career data could be understood. Researches by Bian et al. (2020) demonstrated that contextual embeddings from pretrained models significantly improved job-skill alignment tasks compared to traditional TF-IDF or Word2Vec approaches. However, these models were predominantly used for feature extraction rather than full conversational advisory systems.

The concept of autonomous agents built on top of LLMs gained significant traction following works such as ReAct (Yao et al., 2022) and AutoGPT. These frameworks demonstrated that LLMs could iteratively plan, reason, and execute multi-step tasks by interacting with external tools and knowledge bases. This paradigm is highly relevant to career advising, where the agent must gather user profile data, query job market sources, identify skill gaps, and synthesize a structured roadmap, all in an autonomous chain of reasoning.

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020), further strengthened the ability of LLMs to answer domain-specific queries by grounding their responses in retrieved external knowledge. In the context of career guidance, RAG enables the system to retrieve up-to-date job descriptions, salary benchmarks, and industry trends before generating recommendations, mitigating the hallucination problem commonly observed in standalone LLMs.

Research in personality-aware recommendation systems, particularly those incorporating the Big Five personality model (OCEAN), has shown that incorporating psychological traits into career matching significantly improves user satisfaction and long-term career alignment. Park et al. (2021) demonstrated that personality-augmented recommendation engines outperformed skill-only systems in predicting job satisfaction outcomes. This motivates the inclusion of personality assessment as a core input dimension in the proposed system.

Overall, the literature reveals that while individual components such as LLMs, RAG, and agentic planning have been studied in isolation, no existing system comprehensively integrates all these elements into a unified, conversational, and explainable career advisory platform. This gap forms the primary motivation for the proposed project.

4. Limitations of the Existing Work (Gap Analysis) and Motivation

A critical review of existing career guidance systems and related research reveals several significant limitations that the proposed Intelligent Career Path Advisory Agent aims to address.

4.1 Lack of Holistic User Profiling

Most existing career platforms such as LinkedIn, Naukri, and Glassdoor primarily focus on academic qualifications and technical skills while largely ignoring personality traits, soft skills, extracurricular achievements, and long-term career aspirations. This one-dimensional approach results in recommendations that may be technically relevant but misaligned with a user's actual potential, interests, and values, leading to poor career satisfaction over time.

4.2 Absence of Explainability in Recommendations

Existing career recommendation systems largely function as black-box engines. Users receive a list of suggested job roles or courses without any reasoning or justification. This lack of transparency makes it difficult for users to understand why a particular path was recommended, how their profile maps to industry expectations, or what specific steps they need to take to achieve their goals. Explainability is a critical requirement for building user trust in AI-driven advisory systems.

4.3 Static and Non-Adaptive Guidance

Traditional career counseling tools and even modern platforms offer static assessments, such as one-time aptitude tests or fixed skill assessments, without the ability to adapt to a user's evolving profile or changing market conditions. Real career planning is an iterative, ongoing process. The inability of existing systems to engage in multi-turn dialogue or refine recommendations based on user feedback severely limits their practical utility.

4.4 Disconnect from Real-Time Job Market Dynamics

Career guidance systems built on static datasets quickly become outdated in fast-moving industries such as artificial intelligence, cloud computing, and cybersecurity. Most existing tools do not incorporate real-time job market signals such as emerging skill demands, salary trends, or role availability. This disconnect leads to recommendations that may have been relevant a few years ago but no longer reflect the current professional landscape.

4.5 No Structured Skill Gap Analysis or Learning Roadmap

While some platforms suggest online courses, they rarely provide a structured analysis of the specific gaps between a user's current skill set and the requirements of their target role. A meaningful advisory system should be able to identify precise deficiencies and prescribe a sequenced, prioritized learning plan, including relevant certifications, projects, and timelines, rather than offering a generic list of courses.

4.6 Motivation

The limitations identified above collectively underscore the need for an intelligent, conversational, and explainable career advisory system that is grounded in real-time data and capable of reasoning over complex, multi-dimensional user profiles. The Intelligent Career Path Advisory Agent is motivated by the opportunity to bring together the power of Large Language Models, agentic planning, and retrieval-augmented generation to create a system that not only recommends but also mentors, explains, and guides users toward informed and personalized career decisions. By democratizing access to high-quality career mentorship through an AI-powered agent, this project aims to significantly reduce the career decision-making uncertainty faced by students and early professionals.

5. Functional Requirements for the Prototype (Major Project-Phase 1)

The prototype of the Intelligent Career Path Advisory Agent focuses on demonstrating the core feasibility of the system. The following functional requirements define the key capabilities that the prototype must deliver within the scope of Major Project Phase 1.

Req ID	Functional Requirement (SMART Compliant)	SMART Justification	Expected Outcome
FR1	The system shall allow a user to enter their academic background, technical skills, soft skills, and career interests through a structured input form and store the data as a user profile.	**Specific:** Multi-dimensional profile capture. **Measurable:** Profile is stored and retrievable. **Achievable:** Form-based interface. **Relevant:** Core input for recommendations. **Time-Bound:** Implemented in Sem VI prototype.	A complete user profile is created and used as input to the advisory engine.
FR2	The system shall analyze the user profile using an LLM-powered reasoning engine and generate a list of at least three personalized career path recommendations with explanations for each suggestion.	**Specific:** LLM-based career recommendations. **Measurable:** At least 3 recommendations with reasoning. **Achievable:** Using OpenAI/Gemini API. **Relevant:** Core advisory function. **Time-Bound:** Delivered within prototype scope.	User receives personalized, explained career recommendations rather than generic job suggestions.
FR3	The system shall perform a skill gap analysis by comparing the user's current skill set against the requirements of each recommended career path and present a prioritized learning roadmap with specific resources.	**Specific:** Gap analysis and learning roadmap. **Measurable:** Gaps identified per career path. **Achievable:** LLM + RAG pipeline. **Relevant:** Bridges user to career goals. **Time-Bound:** Implemented in Sem VI.	User receives a structured, actionable plan showing which skills to acquire and in what order.

6. Proposed Methodology for Building Prototype (Major Project-Phase 1)

The prototype of the Intelligent Career Path Advisory Agent will be developed following a structured, modular methodology that integrates user profiling, LLM-based reasoning, retrieval-augmented generation, and a conversational interface. The overall workflow is designed to demonstrate the core feasibility of the system within the scope of Major Project Phase 1.

6.1 System Architecture

The system is organized into the following core modules:

- User Profiling Module: Collects multi-dimensional user data through a structured interface.
- LLM Reasoning Engine: Processes user profiles using a large language model to generate career recommendations with explanations.
- RAG Pipeline: Retrieves real-time job market information from curated knowledge sources to ground recommendations in current data.
- Skill Gap Analyzer: Compares user skills against target role requirements and generates a prioritized learning roadmap.
- Conversational Interface: Provides a natural language chat interface for users to interact with the advisory agent.

6.2 Workflow

The end-to-end data and reasoning flow of the system is as follows:

User Input → Profile Storage → RAG-Augmented Context Retrieval → LLM Reasoning → Career Recommendations + Skill Gap Analysis → Conversational Output

6.3 Software and Tools

The prototype will be implemented using the following technologies:

Component	Technology / Tool
Programming Language	Python 3.11
LLM API	Google Gemini API / OpenAI GPT-4o
Agentic Framework	LangChain / LangGraph
RAG & Vector Store	FAISS / ChromaDB with LangChain
Backend API	FastAPI
Frontend Interface	React.js / Streamlit
Database	SQLite / Firebase (user profiles)

Version Control	Git and GitHub

6.4 Algorithms and Techniques

- Retrieval-Augmented Generation (RAG): Used to retrieve current job role requirements, skill demand data, and industry insights from a curated vector knowledge base before LLM inference.
- Agentic Chain-of-Thought Reasoning: The LLM agent is prompted to decompose career advisory queries into sub-tasks, reason step-by-step, and synthesize a coherent response.
- Semantic Similarity Matching: User skills and job role requirements are encoded as vector embeddings and compared using cosine similarity to identify skill gaps.
- Prompt Engineering: Carefully crafted system prompts guide the LLM to produce structured, explainable career roadmaps tailored to the user profile.

6.5 Development Approach

The prototype will be built iteratively. In the first phase, the user profiling module and basic LLM-powered recommendation engine will be implemented and tested. In the second phase, the RAG pipeline will be integrated to ground recommendations in real-time data. Finally, the skill gap analyzer and conversational interface will be completed, and end-to-end testing will be performed to validate all functional requirements.

7. GitHub Repository Details

The project source code, datasets, experiment notebooks, and documentation are maintained in a GitHub repository to ensure transparency, version control, and collaborative development. As the project is currently in its early phase, the repository will be updated progressively throughout the development lifecycle.

The project repository will be made available at:

https://github.com/nmit-1NT23CS056/final_year_project

The repository will include the implementation of all modules, including the user profiling interface, LLM reasoning engine, RAG pipeline, skill gap analyzer, and the conversational frontend. Supporting resources such as sample user profiles, vector knowledge base setup scripts, and API integration guides will also be included to ensure reproducibility.

8. References

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9. Annexures

9.1 Annexure-1: Definition of Prototype or Proof-Of-Concept

The prototype developed in this project demonstrates the core feasibility of an LLM-powered Intelligent Career Path Advisory Agent. The aim of the prototype is to validate that a user's multi-dimensional profile can be processed by an agentic AI system to generate personalized, explainable, and actionable career recommendations.

The prototype accepts structured user inputs including academic background, technical and soft skills, personality traits, interests, and career goals. These inputs are processed by the LLM reasoning engine, augmented by a retrieval-augmented generation pipeline that draws on a curated knowledge base of job market information. The system generates career path recommendations accompanied by clear reasoning and a structured skill gap analysis pointing to specific learning resources.

While the prototype does not aim to achieve production-level robustness or cover all possible career domains, it successfully demonstrates the technical feasibility and practical value of combining LLMs, agentic planning, and RAG for career advisory purposes. The insights gained from the prototype will directly inform the system architecture and feature set of the final project in the subsequent semester.

9.2 Annexure-2: Definition of Functional Requirements

Functional Requirements describe what the system must do. They define the behaviors, features, and capabilities that the prototype must demonstrate. In this project, functional requirements are written to be SMART compliant, meaning each requirement is Specific, Measurable, Achievable, Relevant, and Time-Bound.

The following table provides the complete SMART-justified functional requirements for the prototype:

Req ID	Functional Requirement (SMART)	SMART Justification	Expected Outcome
FR1	The system shall allow a user to enter their academic background, technical skills, soft skills, and career interests through a structured input form and store the data as a user profile.	**Specific:** Profile capture. **Measurable:** Data stored. **Achievable:** Form UI. **Relevant:** Core input. **Time-Bound:** Sem VI.	A complete user profile is created and used as input to the advisory engine for generating recommendations.
FR2	The system shall analyze the user profile using an LLM-powered reasoning engine and	**Specific:** LLM recommendations. **Measurable:** 3+ paths with	User receives personalized, explained career recommendations

	generate at least three personalized career path recommendations with clear explanations for each suggestion.	reasoning. **Achievable:** Gemini/GPT API. **Relevant:** Core function. **Time-Bound:** Prototype scope.	rather than generic job suggestions.
FR3	The system shall identify skill gaps between the user's profile and each recommended career role and present a structured, prioritized learning roadmap with specific resources.	**Specific:** Gap analysis. **Measurable:** Gaps per role. **Achievable:** LLM + RAG. **Relevant:** Actionable guidance. **Time-Bound:** Sem VI.	User receives a structured, actionable plan showing which skills to acquire and in what order to reach their target career.